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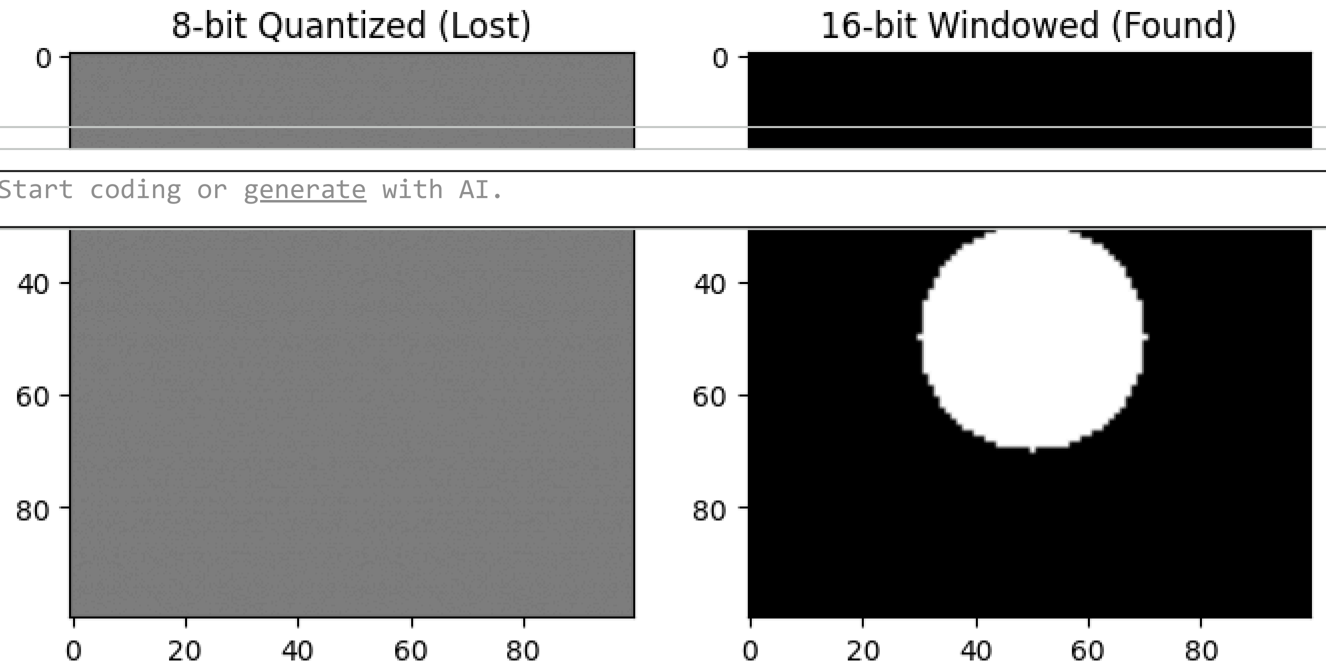
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                              # Import Numpy for mathematical matrix o
import matplotlib.pyplot as plt                                 # Import Matplotlib to visualize the out

# Step 1: Simulate a 16-bit medical scan (range 0-65535)
medical_16bit = np.full((100, 100), 32000, dtype=np.uint16)  # Create a solid gray image array filled
# Introduce a tiny tissue density anomaly (+10 intensity), invisible in 8-bit
cv2.circle(medical_16bit, (50, 50), 20, 32010, -1)  # Draw a filled or outlined circle on the image

# Step 2: Simulate destructive quantization by downcasting to 8-bit
medical_8bit = (medical_16bit / 256).astype(np.uint8)

# Step 3: Mathematically normalize the 16-bit slice to map the subtle data to a visible range
visual_enhancement = cv2.normalize(medical_16bit, None, 0, 255, cv2.NORM_MINMAX, dtype=cv2.CV_8U)  #

# Step 4: Display the medical imaging necessity for high quantization
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(medical_8bit, cmap='gray', vmin=0, vmax=255), plt.title('8-bit Quantized
plt.subplot(122), plt.imshow(visual_enhancement, cmap='gray'), plt.title('16-bit Windowed (Found)')
plt.show()  # Show the final plot window to the user
```



Start coding or generate with AI.